



BAULKHAM HILLS HIGH SCHOOL

TRIAL 2014
YEAR 12 TASK 4

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 180 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-16
- Marks may be deducted for careless or badly arranged work

Total marks – 100

Exam consists of 11 pages.

This paper consists of TWO sections.

Section 1 – Page 2-4 (10 marks)
Questions 1-10

- Attempt Question 1-10
- Allow about **15** minutes for this section

Section II – Pages 5-10 (90 marks)

- Attempt questions 11-16
- Allow about **2 hours and 45** minutes for this section

Table of Standard Integrals is on page 11

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. If $3\sqrt{5} + \sqrt{20} = \sqrt{a}$ then a is

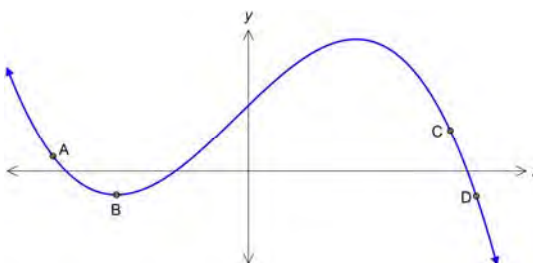
- (A) 5 (B) 25 (C) 125 (D) 15

2. What is the exact value of $\cos \frac{7\pi}{6}$

- (A) $\frac{\sqrt{3}}{2}$ (B) $-\frac{\sqrt{3}}{2}$ (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

3. State which point on the diagram relates to the following

$$y > 0, \frac{dy}{dx} < 0, \frac{d^2y}{dx^2} > 0.$$



- (A) A (B) B (C) C (D) D

4. If $\int_1^4 f(x)dx = 2$ then $\int_1^4 (2f(x) + 3) dx$ is equal to

- (A) 2 (B) 13 (C) 7 (D) 10

5. The $\lim_{\theta \rightarrow 0} \frac{\sin \frac{\theta}{4}}{\frac{\theta}{4}} =$

- (A) 1 (B) 4 (C) $\frac{1}{4}$ (D) 0

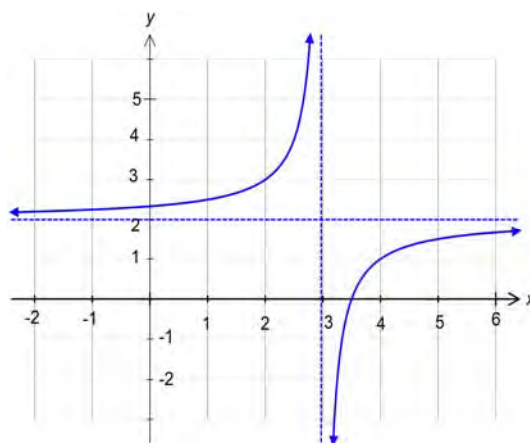
6. The function $f(x) = -3 \sin\left(\frac{\pi x}{5}\right)$ has a period of

- (A) 3 (B) 10 (C) $\frac{\pi}{5}$ (D) $\frac{\pi}{10}$

7. The sum of the series $\frac{4}{9} + \frac{2}{9} + \frac{1}{9} + \frac{1}{18} + \frac{1}{36} + \frac{1}{72} \dots \frac{1}{144}$ is

- (A) $\frac{127}{36}$ (B) $\frac{127}{81}$ (C) $\frac{127}{144}$ (D) $\frac{81}{127}$

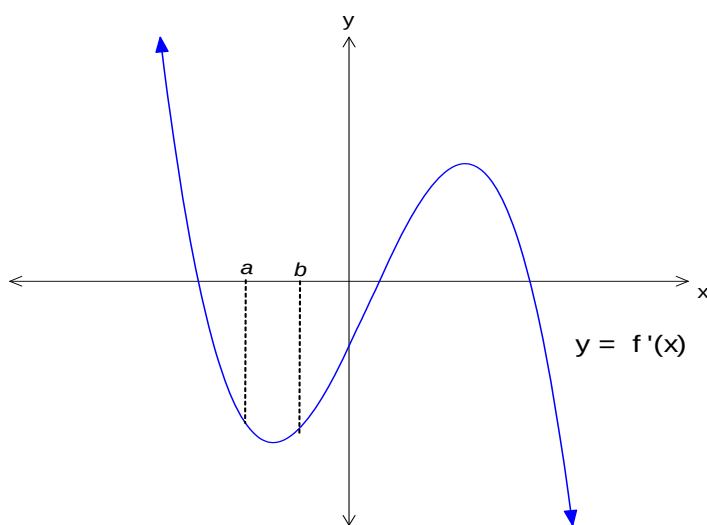
8. Part of the function $y = \frac{a}{x+b} + c$ is shown



The value of a, b and c respectively are

- (A) 2, 3, 2 (B) 2, -3, 2 (C) -2, -3, 2 (D) -2, 3, 2

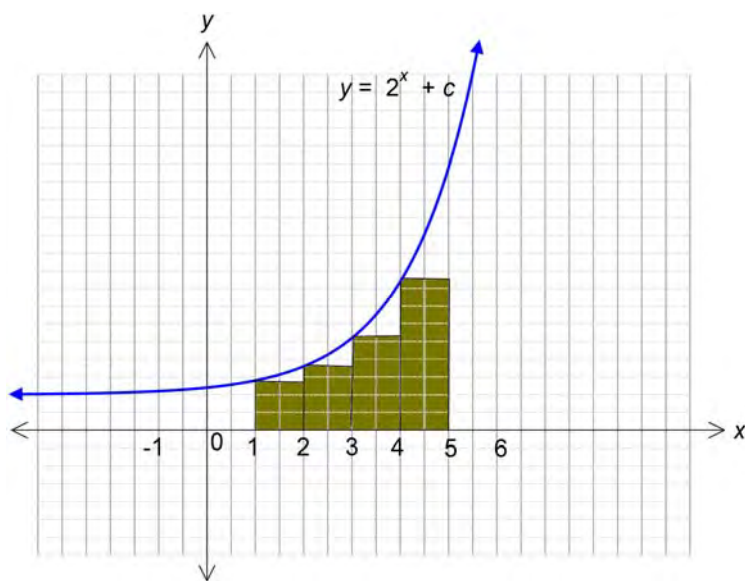
9. The function $y = f'(x)$ shown below



Between $x = a$ and $x = b$, the function $y = f(x)$ will have a

- (A) negative gradient (B) positive gradient
(C) local minimum value (D) local maximum value

10. Consider the graph of $y = 2^x + c$, where c is a real number. The area of the shaded rectangles is used to find an approximation to the area of the region that is bounded by the graph, the x – axis and the line $x = 1$ and $x = 5$



If the total area of the shaded rectangles is 44, then the value of c is

- (A) 14 (B) -4
(C) $\frac{14}{5}$ (D) $\frac{7}{2}$

End of Section 1

Section II – Extended Response**Attempt questions 11-16.****All necessary working should be shown in every question.****Question 11 (15 marks) (Answer on the appropriate page in your booklet)**

a) Solve $\frac{1}{x} = x - 1$ leaving your answer in the exact form.

2

b) Solve $|3x - 4| \leq 8$

2

c) If $\tan\theta = \frac{7}{8}$ and $\cos\theta < 0$, find the exact value of $\sin\theta$.

2

d) Solve for x :

$$9^{2x-3} = 27$$

2

e) Differentiate with respect to x

(i) $y = \cos(x^2)$

2

(ii) $y = \frac{(1-e^{2x})}{x^3}$

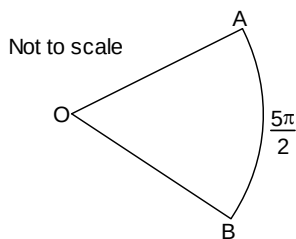
2

f) Find the equation of the normal to $y = 2x^3 - 3x^2$ at a point $x = -1$

3***End of Question 11***

Question 12 (15 marks) (Answer on the appropriate page in your booklet)

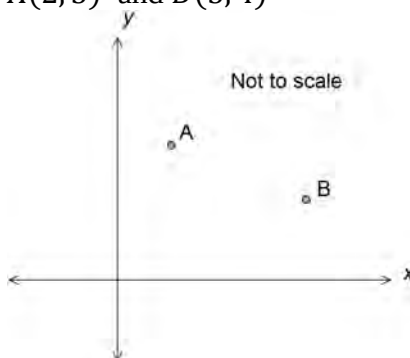
a)



In the diagram AB is an arc of a circle with the centre O . The length of the arc AB is $\frac{5\pi}{2}$ cm. The area of the sector AOB is 4π cm². Find the radius of the sector.

2

b) The diagram shows the point $A(2, 5)$ and $B(5, 4)$



(i) Show that equation of line AB is $x + 3y - 17 = 0$

2

(ii) Find the coordinates of M the midpoint of AB .

1

(iii) Show that the equation of the perpendicular bisector of AB is

2

$$3x - y - 6 = 0$$

(iv) The perpendicular bisector of AB cuts the x -axis at C . Find the coordinates of C .

1

(v) Find the area of $\triangle ABC$

2

c) Give the exact value of $\log_3 \left(\frac{1}{\sqrt{3}} \right)$

1

d) Find $\int \sin \frac{x}{2} dx$.

2

e) If the limiting sum of the series $1 - 2p + 4p^2 \dots$ is $\frac{4}{7}$. Find the value of p .

2

End of Question 12

Question 13 (15 marks) (Answer on the appropriate page in your booklet)

a) Consider the function given by $f(x) = 2x^3 - 3x^2 - 36x + 26$.

(i) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature.

3

(ii) Find the coordinates of any point of inflection.

2

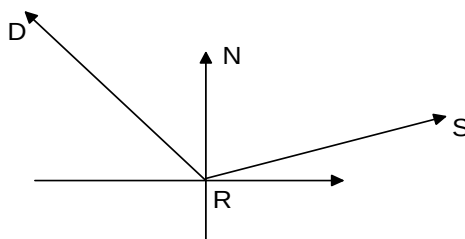
(iii) Hence sketch the graph of $f(x) = 2x^3 - 3x^2 - 36x + 26$ by showing the above information.

2

(iv) For what values of x is the curve concave down and decreasing?

2

b) Danny and Stella are pilots of two light planes which leave Rodeo Station at the same time. Danny flies on a bearing of 330°T at a speed of 180km/h and Stella flies on a bearing of 080°T at a speed of 240km/h. Copy the diagram below onto your answer page and mark the information on the diagram.



(i) How far apart are Danny and Stella after 2 hours?

2

(ii) What is the bearing of Stella from Danny after 2 hours? (to the nearest degree)

2

c) There is one red and three green jelly beans in a jar. One jellybean is selected at random, eaten, and then a second jellybean is selected at random and is also eaten. Find the probability:

(i) The two jellybeans eaten are both green.

1

(ii) The red jellybean is the second one eaten.

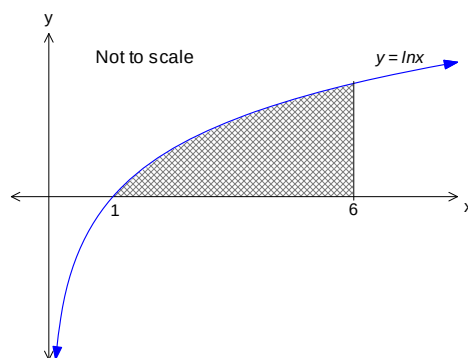
1

End of Question 13

Question 14 (15 marks) (Answer on the appropriate page in your booklet)

- a) The acceleration of a moving body is given by $a = \sqrt{2t+1} \text{ ms}^{-2}$.
If the body starts from rest, find its velocity after 4 seconds. 2

- b) Find the area of the region in the diagram bounded by curve $y = \log x$, x axis and line $x=6$. 2



- c) The velocity of an object is given by the equation $v = 6t - 5 - t^2$ where t is in seconds and velocity v in m/s. It begins its motion at $x = 5$ metres.
- (i) Find an equation for the displacement of the object. 2
- (ii) At what two times is the object stationary? 1
- (iii) Find the distance travelled by the object in first 3 seconds. 2

- d) Consider a parabola $2y = x^2 - 4x$
- (i) Find the coordinates of the focus. 2
- (ii) Find the equation of the directrix. 1

- e) Let A be the point $(-2, 0)$ and $B(6, 0)$. The point $P(x, y)$ is such that $AP \perp PB$.
- (i) Find the gradient of PA . 1
- (ii) Hence find the equation for the locus of P . 2

End of Question 14

Question 15 (15 marks) (Answer on the appropriate page in your booklet)

a) (i) Show that $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = |\tan x|$ 2

(ii) Find the value of x if $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = 1$ for $0 < x < \pi$ 1

b) The population of a species of bacteria P at time t (minutes) grows such that

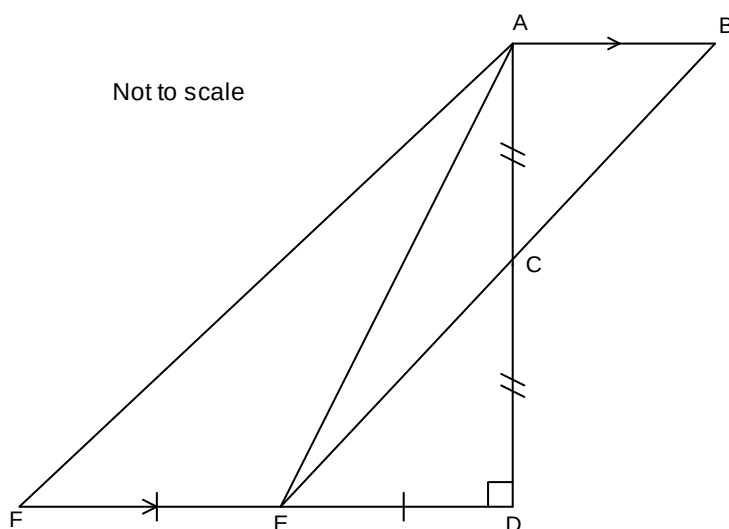
$$P = 2000e^{kt} \text{ where } k \text{ is the positive constant.}$$

(i) Show that the rate of increase of the population is proportional to the size of the population at that time. 1

(ii) Given that the initial population doubles after 4 minutes, calculate the value of k , correct to 3 significant figures. 2

(iii) Find the population after 6 minutes. (correct to nearest whole number) 2

c)



In the diagram $AB \parallel FD$. ADF is a right angled triangle, C is the midpoint of AD and E is the midpoint of FD .

(i) Explain why $\angle CED = \angle ABC$. 1

(ii) Show that $\triangle CDE \equiv \triangle CAB$. 3

(iii) Show that $AF = 2BC$. 3

End of Question 15

Question 16 (15 marks) (Answer on the appropriate page in your booklet)

a) Given that $2\log(x^2y) = 3 + \log x - \log y$. Express y in terms of x .

3

b) Sonia and Jenny want to save a deposit of \$50 000 to buy a house. They devise a savings plan to allow them to achieve this goal. Beginning on 1st January 2014, they deposit \$1000 on the first day of each month into an account which pays 9% p.a compounded monthly.

(i) Find the amount they have in the account on 31st January 2014.
(i.e. at the end of the first month)

1

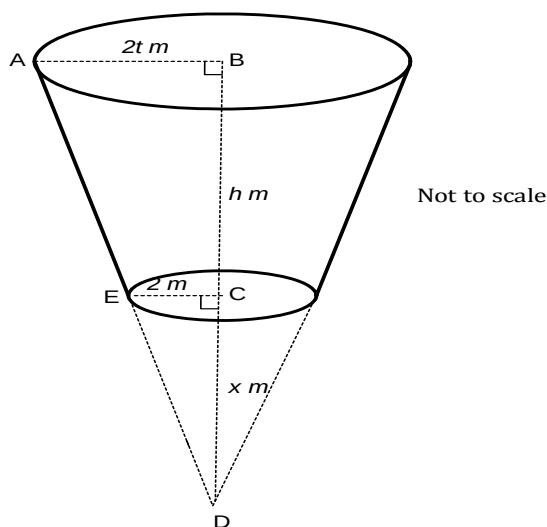
(ii) Show that the amount they have in the account at the end of n months is given by $A = \frac{1007.5(1.0075^n - 1)}{0.0075}$

2

(iii) Hence find the least number of months they need to save their deposit.

2

c) A truncated cone is to be used as a part of a hopper for a grain harvester. It has a height of h metres. The radius AB is to be t times greater than the bottom radius EC which is 2 metres.



(i) If x is the height of the removed section of the original cone, show that

2

$$x = \frac{h}{t-1}$$

(ii) Show that the volume of the truncated cone is given by

2

$$V = \frac{4\pi h}{3} (t^2 + t + 1)$$

(iii) If the upper radius plus the lower radius plus the height of the truncated cone must total 12 metres, calculate the maximum volume of the hopper.

3

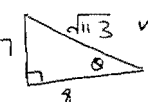
END OF EXAM

MC

1. C 7. C
2. B 8. C
3. A 9. A
4. B 10. D
5. C
6. B

Q11a $\frac{1}{x} = x-1$
 $x^2 - x = 1$ ✓
 $x^2 - x - 1 = 0$
 $x = \frac{1 \pm \sqrt{5}}{2}$ ✓

b $3x-4 \leq 8$ $-6(x-4) \leq 8$
 $3x \leq 12$ $3x-4 \geq -8$ ✓
 $x \leq 4$ $3x \geq -4$
 $x \geq -\frac{4}{3}$
 $-\frac{4}{3} \leq x \leq 4$ ✓

c  $\sin \theta = \frac{\sqrt{7}}{\sqrt{113}}$

d $(3^2)^{2x-3} = 3^3$ ✓
 $4x-6 = 3$

e $1-2p+4p^2 \dots = \frac{4}{7}$
 $\frac{4}{7} = \frac{1}{1+2p}$ ✓
 $4+8p=7$
 $p = \frac{3}{8}$ ✓

Q13a $f(x) = 2x^3 - 3x^2 - 36x + 26$
 $f'(x) = 6x^2 - 6x - 36$
 For stat pts $6x^2 - 6x - 36 = 0$
 $x^2 - x - 6 = 0$
 $x = 3, -2$

∴ coordinates of stat pts are $(3, -55)$ & $(-2, 70)$

$\frac{d^2y}{dx^2} = f''(x) = 12x - 6$
 at $x=3$ $\frac{d^2y}{dx^2} = 36-6 = 30 > 0$ ✓
 ∴ min at $(3, -55)$
 at $x=-2$
 $\frac{d^2y}{dx^2} = -24-6 = -30 < 0$ ∴ max at $(-2, 70)$ ✓

(iv) For possible P.O.I $\frac{d^2y}{dx^2} = 0$
 $12x-6=0 \Rightarrow x = \frac{1}{2}$ ✓

x	-1	$\frac{1}{2}$	1
$\frac{d^2y}{dx^2}$	-18	0	6

∴ There is change in concavity at $x = \frac{1}{2}$
 $(\frac{1}{2}, 4.5)$ is P.O.I ✓
 (must show change in concavity)

Solution

$4x = 9$
 $x = \frac{9}{4}$ ✓
 e. (i) $y' = -\sin x^2 \cdot x \cdot 2x$
 (ii) $y' = \frac{x^2 \cdot -2e^{2x} - 3x^2(1-e^{2x})}{x^4}$
 $= -x^2(xe^{2x} - 3 + 3e^{2x})$

f. $y = 2x^3 - 3x^2$
 $y' = 6x^2 - 6x$
 $m_T = 12$
 $m_N = -\frac{1}{12}$
 $x=1, y=-5$
 ∴ eq of normal is $y+5 = -\frac{1}{12}(x+1)$ ✓
 $x+12y+61=0$

Q12a $\frac{1}{2}r^2\theta = 4\pi$
 $r^2\theta = 8\pi$ — (1)
 $\theta = 5\pi/2$ — (2)
 Divide (1) by (2)
 $r = \frac{8\pi}{5\pi/2} = \frac{16}{5} \text{ cm.}$ ✓

b. (i) $m_{AB} = \frac{4-5}{5-2} = -\frac{1}{3}$ ✓
 eq of AB is $y-5 = -\frac{1}{3}(x-2)$ ✓
 $3y-15 = -x+2$
 $x+3y-17=0$
 (OR if students doing sub., they have to sub. both the pts to get 2 marks)

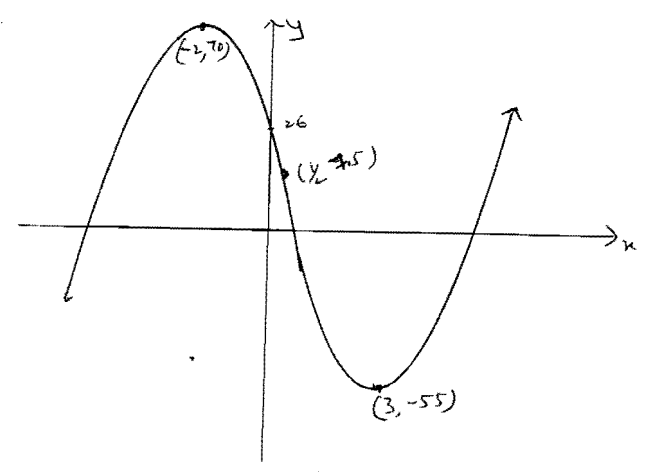
(ii) $m_{AB} = (\frac{7}{2}, \frac{9}{2})$ ✓
 (iii) grad. of bisector = 3 ✓
 ∴ eq is $y-\frac{9}{2} = 3(x-\frac{7}{2})$ ✓
 $2y-9 = 3(x-7)$ ✓
 $6x-2y-12=0$
 $3x-y-6=0$

(iv) $C(2,0)$ ✓

(v) $A = \frac{1}{2} \times 3 \times 5$ ✓
 $= 7.5 \text{ unit}^2$ ✓

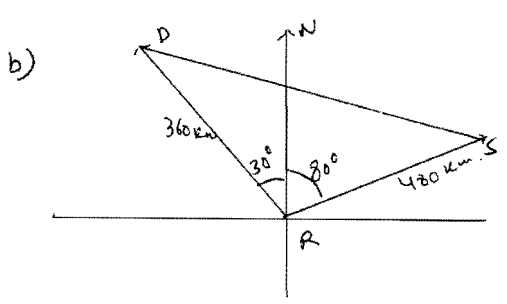
c $\log_3(3)^{-1/2} = -\frac{1}{2}$ ✓

d $\frac{-\cos \frac{x}{2}}{\frac{1}{2}} + C = -2 \cos \frac{x}{2} + C$



(iv) for decreasing fn then $f'(x) < 0$

$6x^2 - 6x - 36 < 0$
 $(x-3)(x+2) < 0$
 $-2 < x < 3$
 For concave down $12x-6 < 0$
 $x < \frac{1}{2}$
 ∴ $-2 < x < \frac{1}{2}$ ✓



$$\angle ORS = 30^\circ + 80^\circ = 110^\circ \quad \checkmark$$

$$DS^2 = OR^2 + RS^2 - 2 \times OR \times RS \cos 110^\circ$$

$$= 360^2 + 480^2 - 2 \times 360 \times 480 \times \cos 110^\circ \quad \checkmark$$

$$= 478202.16 \dots$$

$$DS = 692 \text{ to the nearest km.}$$

$$(691.5)$$

$$\frac{\sin Q}{480} = \frac{\sin 110^\circ}{691.52} \quad \checkmark$$

$$\sin Q = \frac{480 \times \sin 110^\circ}{691.52} \dots$$

$$= 0.65226 \dots$$

$$Q = 40.712 \dots$$

$$= 41^\circ$$

$$\text{Bearing} = 180 - 41 - 30 \quad \checkmark$$

$$= 109^\circ T$$

$$\text{c) i) } P(6, 6) = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2} \quad \checkmark$$

$$\text{(ii) } P(6, 8) = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4} \quad \checkmark$$

$$\text{Q14 } a = \sqrt{2t+1}$$

$$\frac{a}{t} = 0, \quad v \text{ at } t=4$$

$$v=0$$

$$v = \frac{d}{dt} (2t+1)^{3/2} + C$$

$$= \frac{(2t+1)^{3/2}}{3} + C$$

$$\text{at } t=0, v=0$$

$$0 = \frac{1}{3} + C$$

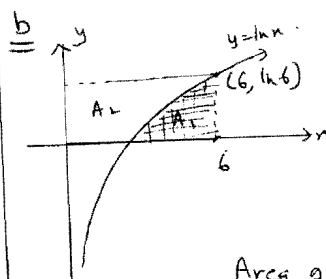
$$C = -\frac{1}{3}$$

$$v = \frac{(2t+1)^{3/2}}{3} - \frac{1}{3}$$

$$\text{at } t=4$$

$$v = \frac{(9)^{3/2}}{3} - \frac{1}{3}$$

$$= 9 - \frac{1}{3} = \frac{26}{3} \text{ m/sec to the right} \quad \checkmark$$



$$y = \log_2 x$$

$$x^y = x$$

$$A_2 = \int_0^{\ln 6} e^y dy$$

$$= e^y \Big|_0^{\ln 6} = 6 - 1 = 5$$

$$\text{Area of rectangle} = \ln 6 \times 6 = 10.7505 \dots$$

$$A_1 = \text{Area} - A_2$$

$$= 10.7505 - 5$$

$$= 5.75 \text{ unit}^2$$

$$\text{c) } v = 6t - 5 - t^2$$

$$\text{i) } x = \frac{6t^2}{2} - 5t - \frac{t^3}{3} + C \quad \checkmark$$

$$\text{at } t=0, x=5$$

$$5 = C \quad \checkmark$$

$$x = 3t^2 - \frac{t^3}{3} - 5t + 5$$

$$\text{(ii) Stationary when } v=0$$

$$6t - 5 - t^2 = 0$$

$$(t-5)(t-1) = 0$$

$$t = 1, 5 \text{ sec.} \quad \checkmark$$

$$\text{(iii) dist} = \left| \int_0^1 6t - 5 - t^2 dt \right| + \left| \int_1^5 6t - 5 - t^2 dt \right|$$

$$= \left| 3t^2 - 5t - \frac{t^3}{3} \Big|_0^1 \right| + \left| 3t^2 - 5t - \frac{t^3}{3} \Big|_1^5 \right|$$

$$= \left| (3 - 5 - \frac{1}{3} - 0) \right| + \left| (27 - 15 - 9) - (12 - 10 - \frac{8}{3}) \right|$$

$$= 3 + (3 - \frac{2}{3}) = 6\frac{2}{3} \text{ m.} \quad \checkmark$$

$$\text{d) } 2y = x^2 - 4x$$

$$x^2 - 4x + 4 = 2y + 4$$

$$(x-2)^2 = 2(y+2) \quad \checkmark$$

$$\text{vertex} = (2, -2) \quad \checkmark$$

$$4a = 2$$

$$a = \frac{1}{2}$$

$$\text{(i) Focus } (2, -\frac{1}{2}) \quad \checkmark$$

$$\text{in direction is } y = -2x \quad \checkmark$$

$$\text{e) (i) } m_{PA} = \frac{y-0}{x+2} = \frac{y}{x+2} \quad \checkmark$$

$$\text{(ii) } m_{PB} = \frac{y-0}{x-6} = \frac{y}{x-6}$$

$$\text{Since } AP \perp PB$$

$$\therefore m_{PA} \times m_{PB} = -1 \quad \checkmark$$

$$\frac{y}{x+2} \times \frac{y}{x-6} = -1$$

$$y^2 = -(x+2)(x-6)$$

$$x^2 - 4x + y^2 = 12 \quad \checkmark \quad \text{or } (x-2)^2 + y^2 = 16$$

$$\text{Q15 a(i) LHS}$$

$$\frac{\csc^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}$$

$$= \frac{1 + \cot^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}$$

$$= \frac{1 - \cos^2 x}{\cos^2 x} = \frac{\sin^2 x}{\cos^2 x} = \frac{\sin x}{\cos x} = \tan x = \text{RHS.}$$

$$\text{(ii) } \tan x = 1 \quad \therefore x = \pi/4 \quad \checkmark$$

$$\text{b) } P = 2000e^{kt}$$

$$\text{(i) } \frac{dP}{dt} = k \times 2000e^{kt}$$

$$\frac{dP}{dt} = kP \quad \therefore \text{proportional} \quad \checkmark$$

$$\text{(ii) } t=0 \quad P = 2000e^0 = 2000$$

$$t=4 \quad P = 4000 \quad \checkmark$$

$$\therefore 4000 = 2000e^{4k}$$

$$2 = e^{4k}$$

$$\ln 2 = 4k \ln e$$

$$k = \frac{\ln 2}{4} = 0.1732 \dots$$

$$= 0.173 \quad \checkmark$$

ii) $t = 6$

$$P = 2000e^{6k} \quad \checkmark$$

$$= 5626.62$$

$$= 5627 \quad \checkmark$$

Q10 $\angle CED = \angle ABC$
(alternate $\angle$ on $AB \parallel FD$) $\checkmark$

(ii) In $\triangle CDE$ & $\triangle ACB$
 $\angle CED = \angle ABC$ proved above
 $\angle CDE = \angle CAB$ (alternate $\angle$ on $AB \parallel FD$) $\checkmark$
 $AC = CD$ (given) $\checkmark$
 $\therefore \triangle CDE \equiv \triangle CAB$ (AAS) $\checkmark$

(iii) $AB = ED$ (matching sides of $\equiv$ triangles)
 $EO = EF$ (E is the mid pt of FD) $\checkmark$
 $\therefore AB = EF$ & also $AB \parallel EF$
 $\therefore ABFE$ is a parallelogram $\checkmark$

$$= 1000 \left[\frac{1.0075(1.0075^n - 1)}{1.0075 - 1} \right] \quad \checkmark$$

$$= 1007.50 \left(\frac{1.0075^n - 1}{0.0075} \right)$$

(iii) $50000 = 1007.5 \left(\frac{1.0075^n - 1}{0.0075} \right)$

$$0.3722 = 1.0075^n - 1$$

$$1.3722 = 1.0075^n$$

$$\log 1.3722 = n \log 1.0075 \quad \checkmark$$

$$n = \frac{\log 1.3722}{\log 1.0075}$$

$$n = 42.347$$

$\therefore$ least no. of months = 43 $\checkmark$

Q11 In $\triangle ABD$ & $\triangle ECD$
 $\angle ECD = \angle ABD = 90^\circ$ (given)
 $\angle ADB$ is common
 $\triangle ABD \sim \triangle ECD$

$\therefore \frac{x}{2t} = \frac{x}{2+h}$ (Ratio of all matching sides are =)
 $h + 2 = tx$
 $h = 2t - x$
 $x = \frac{h}{1-t}$
 - could use Trig
 but ratio had to be correct

$AF = BE$ (opposite sides of parallelogram)
 But $BE = BC + CE$
 $= 2BC$ (as $BC = CE$ given) $\checkmark$
 $\therefore AF = 2BC$

Q16 a $2 \ln(x^2 y) = 3 + \ln x - \ln y$

$$2 \ln x^2 + 2 \ln y = 3 + \ln x - \ln y$$

$$4 \ln x + 2 \ln y = 3 + \ln x - \ln y$$

$$3 \ln x + 3 \ln y = 3$$

$$\ln x + \ln y = 1$$

$$\ln xy = 1$$

$$xy = e$$

$$y = \frac{e}{x}$$

3 marks correct
answer for correct
working.

2 marks
substantial
log knowledge
1 mark some solid
working

b (i) $A_1 = 1000(1 + \frac{0.09}{12})^1$
 $= 1000(1.0075)$
 $= \$1007.50 \quad \checkmark$ (must calculate)

(ii) $A_2 = 1000(1.0075)^2$
 $A_3 = 1000(1.0075)^3$
 $\vdots$
 $A_n = 1000(1.0075)^n$
 $A = 1000(1.0075) + 1000(1.0075)^2 + \dots + 1000(1.0075)^n$
 $= 1000 \left[1.0075 + 1.0075^2 + \dots + 1.0075^n \right] \quad \checkmark$
 4 p.

(iii) $V_{\text{cone}} = \frac{1}{3} \pi r^2 h$

$V_{\text{truncated cone}} = V_{\text{cone}} - V_{\text{small cone}}$

$$= \frac{1}{3} \pi (2t)^2 (h+t) - \frac{1}{3} \pi 4x$$

$$= \frac{4}{3} \pi (t^2 h + t^2 x - x)$$

$$= \frac{4}{3} \pi \left(t^2 h + \frac{t^2 h}{t-1} - \frac{h}{t-1} \right)$$

$$= \frac{4}{3} \pi h \left(t^2 + \frac{t^2}{t-1} - \frac{1}{t-1} \right) \quad \checkmark$$

$$= \frac{4}{3} \pi h \left(t^2 + \frac{t^2 - 1}{t-1} \right) \quad \checkmark$$

$$= \frac{4}{3} \pi h (t^2 + t + 1)$$

(iii) $2t + 2 + h = 12$

$$2t + h = 10$$

$$h = 10 - 2t \quad \checkmark$$

$$V = \frac{4}{3} \pi (10 - 2t) [t^2 + t + 1]$$

$$= \frac{8\pi}{3} [(5-t)(t^2 + t + 1)]$$

$$= \frac{8\pi}{3} [5t^2 + 5t + 5 - t^3 - t^2 - t]$$

$$= \frac{8\pi}{3} [-t^3 + 4t^2 + 4t + 5]$$

$$\frac{dV}{dt} = \frac{8\pi}{3} [-3t^2 + 8t + 4]$$

For max vol. $\frac{dV}{dt} = 0$

$$-3t^2 + 8t + 4 = 0$$

$$3t^2 - 8t - 4 = 0$$

$$t = \frac{8 \pm \sqrt{112}}{6}$$

$$= \frac{4 \pm 2\sqrt{7}}{3} \quad \checkmark$$

$$\text{But } t = \frac{4 + 2\sqrt{7}}{3} = 3.09$$

$$\text{As } t > 0$$

$$\frac{d^2V}{dt^2} = \frac{8\pi}{3} [-6t + 8]$$

$$= -88.29 < 0$$

✓ Volume
+ testing.

∴ max at $t = 3.09$

$$\text{Hence } V = \underline{218.225 \text{ unit}^3}$$